

Shrimad Mishra

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Senior Software Engineer | Generative AI | Voicebots | Multi-Agent Systems | LangGraph, Kubernetes, AWS, CI/CD

SUMMARY

Senior Software Engineer with ~5 years at Cogno AI / Exotel (Cogno AI was acquired by Exotel) building high-scale, real-time backend systems. Started the company's AI voicebot from scratch and scaled it into a production-grade Go platform using a LangGraph multi-agent architecture, Model Context Protocol (MCP) and Pipecat real-time voice streaming, deployed on containerized multi-cloud infrastructure.

EXPERIENCE

Senior Software Engineer | Exotel (acquired Cogno AI) | Bengaluru, India | Oct 2024 - Present

- Owned the Voicebot end-to-end, scaling it from a Python/Django prototype into a production-grade Go service.
- Redesigned the conversation engine from a single LLM call into a LangGraph multi-agent supervisor architecture for dynamic, context-aware flow control.
- Added Model Context Protocol (MCP) support to standardize tool integration and context fetching for agentic workflows.
- Integrated Pipecat for real-time, low-latency voice streaming and audio pipeline orchestration.
- Built the SIP connector for the voicebot, handling call signaling and real-time media streams.
- Reduced voicebot response latency by adding conversational audio fillers.
- Containerized all services (Docker, Kubernetes) and deployed multi-cloud infrastructure on AWS and Oracle OCI (VPC networking, load balancing, CDN).
- Architected CI/CD pipelines enabling 50+ production deployments per week with zero downtime.
- Engineered full-stack features (React, Java) for the customer-facing voicebot dashboard used to create bots and track their performance, integrated with Keycloak (authentication) and Apache Superset (analytics).
- Received multiple Medallions of Honor for contributions to voicebot and automation infrastructure.

Software Engineer | Cogno AI (acquired by Exotel) | Bengaluru, India | Jan 2022 - Sep 2024

Joined as a contractor in Jan 2022; converted to full-time in Aug 2022.

- Owned the Chatbot and Co-browsing products (Jan 2022 - Jul 2023), reviewing PRDs for technical feasibility, guiding the team on execution, and reviewing solutions and pull requests.
- Built a BERT-based intent recognition and entity extraction engine for core NLU, reducing support overhead by 40%.
- Architected a real-time Co-browse platform using WebSocket (WSS) DOM streaming and a React frontend, supporting websites and Android apps.
- Implemented an early Retrieval-Augmented Generation (RAG) chatbot using document embeddings, improving retrieval relevance over keyword search.
- Drove backend reliability improvements, reducing bugs by 60% and client-reported issues by ~45%.
- Cut load time by 70% through database and backend optimizations.
- Engineered real-time processing pipelines with Apache Airflow, achieving sub-3-second execution latency.
- Built observability and logging infrastructure (OpenSearch, Logstash, Filebeat) for production monitoring.
- Started the Voicebot from scratch in Aug 2023, designing and delivering the initial MVP in Python/Django.

Machine Learning Engineer Intern | Applex.in | Kolkata, India | Jun 2021 - Aug 2021

- Built an image-to-text extraction pipeline for the Snaplingo app, enabling document-to-PDF conversion with 95% extraction accuracy.
- Designed a computer-vision table detection module converting 100+ unstructured document images daily into structured Excel files with 90%+ parsing accuracy.

SKILLS

Languages: Go, Python, Java, JavaScript, C++

AI & Agentic Systems: LangGraph (multi-agent), LangChain, Model Context Protocol (MCP), RAG, Vector Embeddings, BERT, Azure OpenAI, Prompt Engineering, N8N

Voice & Real-Time: Pipecat, SIP, WebSockets (WSS)

Backend & Frontend: Django, Flask, REST APIs, Microservices, Apache Kafka, Nginx, React

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, RDS, S3, VPC), Oracle OCI, Azure, CI/CD, Jenkins, CDN

Data & Observability: PostgreSQL, Redis, OpenSearch, Logstash, Filebeat, Apache Airflow, Apache Superset, Keycloak

Tools: Git, Jira, Linux

EDUCATION

Bachelor of Engineering (B.E.), Computer Science and Engineering | Dr. Ambedkar Institute of Technology | Bengaluru, India | Aug 2018 - Jun 2022

- CGPA: 8.8/10 | Coursework: Data Structures, Algorithms, Machine Learning, Deep Learning, Databases
- Publication: "Prediction of Knee-Replacement using Deep-Learning Approach", IEEE, Oct 2022.